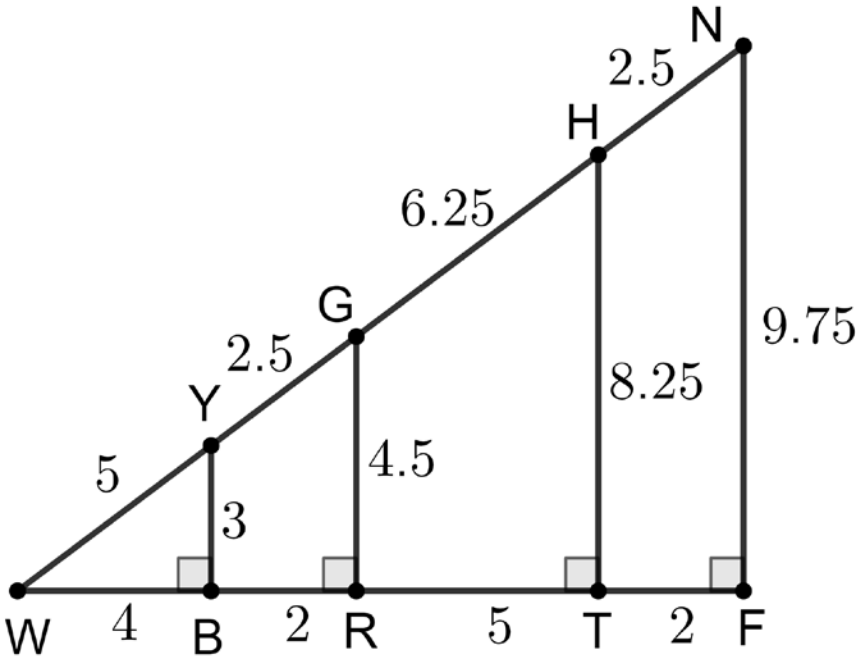
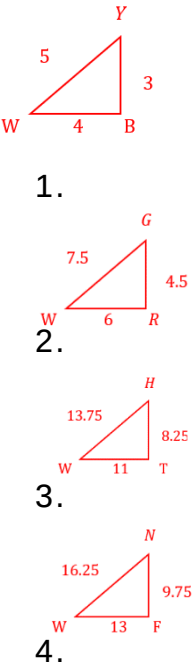


The diagram shows measurements for $\triangle YBW$, $\triangle GRW$, $\triangle HTW$, and $\triangle NFW$.



Complete the table using $\angle W$ as the reference angle.



Triangle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
$\triangle YBW$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{3}{4}$
$\triangle GRW$	$\frac{4.5}{7.5}$	$\frac{6}{7.5}$	$\frac{4.5}{6}$
$\triangle HTW$	$\frac{8.25}{13.75}$	$\frac{11}{13.75}$	$\frac{8.25}{11}$
$\triangle NFW$	$\frac{9.75}{16.25}$	$\frac{13}{16.25}$	$\frac{9.75}{13}$

For $\triangle YBW$, $\triangle GRW$, $\triangle HTW$, and $\triangle NFW$ find the decimal equivalent for each ratio.

	Ratio	$\triangle YBW$	$\triangle GRW$	$\triangle HTW$	$\triangle NFW$
5.	$\frac{\text{opposite}}{\text{hypotenuse}}$	0.6	0.6	0.6	0.6
6.	$\frac{\text{adjacent}}{\text{hypotenuse}}$	0.8	0.8	0.8	0.8
7.	$\frac{\text{opposite}}{\text{adjacent}}$	0.75	0.75	0.75	0.75

Complete the statements.

8. The decimal equivalents of the

- ☐ $\frac{\text{opposite}}{\text{adjacent}}$
- ☐ $\frac{\text{adjacent}}{\text{hypotenuse}}$
- ☐ $\frac{\text{opposite}}{\text{hypotenuse}}$
- ☒ three

ratios for triangles YBW ,

GRW , HTW , and NFW are equal.

9. Because triangles YBW , GRW , HTW , and NFW are

- ☐ congruent
- ☒ similar

triangles,

their side lengths are

- ☐ congruent.
- ☒ proportional.

10. Since triangles YBW , GRW , HTW , and NFW share

- ☒ $\angle W$
- ☐ side $\overline{FN}$
- ☐ sides $\overline{FN}$ and $\overline{RN}$

and

are

- ☐ acute
- ☐ obtuse
- ☒ right

triangles, it is possible to define trigonometric ratios in any right

triangle.